

Supporting Information

Interpretable Machine Learning for Predicting GHS Chemical Hazard Classifications: A Multi-Label Approach Using 243,323 Compounds and Scaffold-Based Validation

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The tables below are also provided as an Excel workbook, *publication_supplementary_tables.xlsx*, in which every table appears in full. The underlying dataset, descriptors and trained models are archived at <https://doi.org/10.5281/zenodo.21876611> and the analysis code at <https://doi.org/10.5281/zenodo.21903565>.

Table S0

GHS label schema: column names, pictogram codes, their authoritative United Nations meanings, and the correspondence with the names used in the original study design.

column name	pictogram code	authoritative UN PubChem meaning	name in original proposal	was renamed
GHS01_Explosive	GHS01	Explosive (exploding bomb)	GHS01_Explosive	no
GHS02_Flammable	GHS02	Flammable (flame)	GHS02_Flammable	no
GHS03_Oxidising	GHS03	Oxidiser (flame over circle)	GHS03_Oxidising	no
GHS04_CompressedGas	GHS04	Compressed gas (gas cylinder)	GHS04_CompressedGas	no
GHS05_Corrosive	GHS05	Corrosive (corrosion)	GHS05_Corrosive	no
GHS06_AcuteToxicity	GHS06	Acute toxicity (skull and crossbones)	GHS06_AcuteToxicity	no
GHS07_Irritant	GHS07	Irritant / harmful (exclamation mark)	GHS07_HealthHazard	YES
GHS08_HealthHazard	GHS08	Serious health hazard (health hazard)	GHS08_Environmental	YES
GHS09_Environmental	GHS09	Environmental hazard (environment)	GHS09_Irritant	YES

9 rows. Source sheet: S0_label_schema.

Table S1

Full dataset statistics after cleaning: compound counts, percentages and imbalance ratios per hazard class.

GHS Column	Pictogram Code	Actual Meaning	N Positive	N Negative	Percent of Dataset	Imbalance Ratio NegPerPos	Name in original proposal	Note
GHS01_Explosive	GHS01	Explosive (exploding bomb)	184	243139	0.076	1321.41	GHS01_Explosive	
GHS02_Flammable	GHS02	Flammable (flame)	10632	232691	4.37	21.89	GHS02_Flammable	
GHS03_Oxidising	GHS03	Oxidiser (flame over circle)	419	242904	0.172	579.72	GHS03_Oxidising	
GHS04_CompressedGas	GHS04	Compressed gas (gas cylinder)	285	243038	0.117	852.76	GHS04_CompressedGas	
GHS05_Corrosive	GHS05	Corrosive (corrosion)	43336	199987	17.81	4.61	GHS05_Corrosive	
GHS06_AcuteToxicity	GHS06	Acute toxicity (skull and crossbones)	15201	228122	6.247	15.01	GHS06_AcuteToxicity	
GHS07_Irritant	GHS07	Irritant / harmful (exclamation mark)	220020	23303	90.423	0.11	GHS07_HealthHazard	Renamed from the original study design to match the official UN pictogram meaning. The underlying data were bound to the numeric code and are unchanged.
GHS08_HealthHazard	GHS08	Serious health hazard (health hazard)	13606	229717	5.592	16.88	GHS08_Environmental	Renamed from the original study design to match the official UN pictogram meaning. The underlying data were bound to the numeric code and are unchanged.
GHS09_Environmental	GHS09	Environmental hazard (environment)	17735	225588	7.289	12.72	GHS09_Irritant	Renamed from the original study design to match the official UN pictogram meaning. The underlying data were bound to the numeric code and are unchanged.

9 rows. Source sheet: S1_dataset_statistics.

Table S2

Complete hyperparameter search results for all three algorithms.

Algorithm	GHS Class	Best CV Score	N Iterations	CV Folds	Best Parameters
RandomForest	all (multi-output)	0.8246	5	3	{"estimator__n_estimators": 500, "estimator__min_samples_split": 5, "estimator__min_samples_leaf": 1, "estimator__max_features": "log2", "estimator__max_depth": 50}
XGBoost	GHS01_Explosive	0.8917	5	3	{"subsample": 0.9, "reg_lambda": 10.0, "reg_alpha": 1.0, "n_estimators": 200, "max_depth": 10, "learning_rate": 0.01, "colsample_bytree": 0.9}
XGBoost	GHS02_Flammable	0.9625	5	3	{"subsample": 0.6, "reg_lambda": 10.0, "reg_alpha": 0.01, "n_estimators": 300, "max_depth": 6, "learning_rate": 0.05, "colsample_bytree": 0.7}
XGBoost	GHS03_Oxidising	0.9879	5	3	{"subsample": 0.6, "reg_lambda": 10.0, "reg_alpha": 0.01, "n_estimators": 300, "max_depth": 6, "learning_rate": 0.05, "colsample_bytree": 0.7}
XGBoost	GHS04_CompressedGas	0.9972	5	3	{"subsample": 0.6, "reg_lambda": 10.0, "reg_alpha": 0.01, "n_estimators": 300, "max_depth": 6, "learning_rate": 0.05, "colsample_bytree": 0.7}
XGBoost	GHS05_Corrosive	0.8565	5	3	{"subsample": 0.6, "reg_lambda": 10.0, "reg_alpha": 0.01, "n_estimators": 300, "max_depth": 6, "learning_rate": 0.05, "colsample_bytree": 0.7}
XGBoost	GHS06_AcuteToxicity	0.7775	5	3	{"subsample": 0.9, "reg_lambda": 10.0, "reg_alpha": 1.0, "n_estimators": 200, "max_depth": 10, "learning_rate": 0.01, "colsample_bytree": 0.9}
XGBoost	GHS07_Irritant	0.7775	5	3	{"subsample": 0.6, "reg_lambda": 10.0, "reg_alpha": 0.01, "n_estimators": 300, "max_depth": 6, "learning_rate": 0.05, "colsample_bytree": 0.7}
XGBoost	GHS08_HealthHazard	0.8831	5	3	{"subsample": 0.6, "reg_lambda": 10.0, "reg_alpha": 0.01, "n_estimators": 300, "max_depth": 6, "learning_rate": 0.05, "colsample_bytree": 0.7}
XGBoost	GHS09_Environmental	0.851	5	3	{"subsample": 0.6, "reg_lambda": 10.0, "reg_alpha": 0.01, "n_estimators": 300, "max_depth": 6, "learning_rate": 0.05, "colsample_bytree": 0.7}
SVM	all (multi-output)	0.745	30	3	{"estimator__svm_kernel": "rbf", "estimator__svm_gamma": 0.01, "estimator__svm_C": 0.1}

11 rows. Source sheet: S2_hyperparameter_search.

Table S3

Full performance metrics: every model, class, metric and decision threshold, with bootstrap confidence intervals.

Model	Pictogram Code	Threshold Type	Threshold Value	N Test Positive	AUC ROC	AUC CI95 lower	AUC CI95 upper	Average Precision	F1	MCC	Precision	Recall
RandomForest	GHS01	default_0.5	0.5	18	0.9722	0.9518	0.9889	0.1184	0.0576	0.1277	0.0304	0.5556
RandomForest	GHS01	calibrated_F1	0.879101	18	0.9722	0.9518	0.9889	0.1184	0.1379	0.1416	0.1818	0.1111
RandomForest	GHS01	safety_recall90	0.152825	18	0.9722	0.9518	0.9889	0.1184	0.0101	0.0659	0.0051	1.0
RandomForest	GHS02	default_0.5	0.5	1063	0.952	0.9457	0.9581	0.6242	0.3864	0.433	0.2456	0.9059
RandomForest	GHS02	calibrated_F1	0.797954	1063	0.952	0.9457	0.9581	0.6242	0.589	0.5702	0.5882	0.5898
RandomForest	GHS02	safety_recall90	0.497792	1063	0.952	0.9457	0.9581	0.6242	0.385	0.4321	0.2443	0.9078
RandomForest	GHS03	default_0.5	0.5	41	0.9321	0.8751	0.978	0.4208	0.1187	0.2038	0.0652	0.6585
RandomForest	GHS03	calibrated_F1	0.93163	41	0.9321	0.8751	0.978	0.4208	0.42	0.4258	0.3559	0.5122
RandomForest	GHS03	safety_recall90	0.205989	41	0.9321	0.8751	0.978	0.4208	0.0393	0.1229	0.0201	0.8293
RandomForest	GHS04	default_0.5	0.5	28	0.9726	0.9102	0.9994	0.5792	0.1289	0.256	0.0691	0.9643
RandomForest	GHS04	calibrated_F1	0.96652	28	0.9726	0.9102	0.9994	0.5792	0.4889	0.5038	0.6471	0.3929
RandomForest	GHS04	safety_recall90	0.851522	28	0.9726	0.9102	0.9994	0.5792	0.4107	0.4733	0.2738	0.8214
RandomForest	GHS05	default_0.5	0.5	4333	0.8448	0.8378	0.8513	0.6571	0.5482	0.4422	0.44	0.7272
RandomForest	GHS05	calibrated_F1	0.561174	4333	0.8448	0.8378	0.8513	0.6571	0.5901	0.4961	0.5568	0.6275
RandomForest	GHS05	safety_recall90	0.351714	4333	0.8448	0.8378	0.8513	0.6571	0.4147	0.2919	0.2685	0.91
RandomForest	GHS06	default_0.5	0.5	1520	0.802	0.7905	0.813	0.2487	0.269	0.2507	0.1687	0.6638
RandomForest	GHS06	calibrated_F1	0.601884	1520	0.802	0.7905	0.813	0.2487	0.3162	0.2669	0.2778	0.3671
RandomForest	GHS06	safety_recall90	0.369084	1520	0.802	0.7905	0.813	0.2487	0.189	0.185	0.106	0.8717
RandomForest	GHS07	default_0.5	0.5	21967	0.8061	0.7978	0.8143	0.9718	0.8518	0.3029	0.9579	0.7668
RandomForest	GHS07	calibrated_F1	0.232387	21967	0.8061	0.7978	0.8143	0.9718	0.9491	0.1748	0.9077	0.9945
RandomForest	GHS07	safety_recall90	0.354999	21967	0.8061	0.7978	0.8143	0.9718	0.9242	0.2804	0.9317	0.9168
RandomForest	GHS08	default_0.5	0.5	1360	0.8645	0.8559	0.8733	0.3053	0.311	0.3109	0.1992	0.7088
RandomForest	GHS08	calibrated_F1	0.686155	1360	0.8645	0.8559	0.8733	0.3053	0.363	0.324	0.3492	0.3779
RandomForest	GHS08	safety_recall90	0.37313	1360	0.8645	0.8559	0.8733	0.3053	0.2326	0.2605	0.1338	0.8875
RandomForest	GHS09	default_0.5	0.5	1773	0.8637	0.8553	0.8721	0.3834	0.3791	0.3644	0.2538	0.749
RandomForest	GHS09	calibrated_F1	0.664859	1773	0.8637	0.8553	0.8721	0.3834	0.4149	0.3662	0.3812	0.4552
RandomForest	GHS09	safety_recall90	0.412591	1773	0.8637	0.8553	0.8721	0.3834	0.3318	0.3325	0.2072	0.8319
XGBoost	GHS01	default_0.5	0.5	18	0.9859	0.9743	0.9953	0.1507	0.0785	0.1713	0.0415	0.7222
XGBoost	GHS01	calibrated_F1	0.912269	18	0.9859	0.9743	0.9953	0.1507	0.1818	0.2355	0.5	0.1111
XGBoost	GHS01	safety_recall90	0.067481	18	0.9859	0.9743	0.9953	0.1507	0.0022	0.019	0.0011	1.0
XGBoost	GHS02	default_0.5	0.5	1063	0.9708	0.9654	0.9751	0.7552	0.5427	0.5638	0.3907	0.8881
XGBoost	GHS02	calibrated_F1	0.891765	1063	0.9708	0.9654	0.9751	0.7552	0.7071	0.694	0.7173	0.6971
XGBoost	GHS02	safety_recall90	0.535326	1063	0.9708	0.9654	0.9751	0.7552	0.5607	0.5776	0.4115	0.8796

Model	Pictogram Code	Threshold Type	Threshold Value	N Test Positive	AUC ROC	AUC CI95 lower	AUC CI95 upper	Average Precision	F1	MCC	Precision	Recall
XGBoost	GHS03	default_0.5	0.5	41	0.9267	0.8617	0.9807	0.5438	0.4355	0.4616	0.3253	0.6585
XGBoost	GHS03	calibrated_F1	0.978498	41	0.9267	0.8617	0.9807	0.5438	0.5797	0.5897	0.7143	0.4878
XGBoost	GHS03	safety_recall90	0.040324	41	0.9267	0.8617	0.9807	0.5438	0.2357	0.3228	0.1396	0.7561
XGBoost	GHS04	default_0.5	0.5	28	0.9951	0.9841	0.9999	0.8092	0.6667	0.6887	0.5319	0.8929
XGBoost	GHS04	calibrated_F1	0.969746	28	0.9951	0.9841	0.9999	0.8092	0.7368	0.7366	0.7241	0.75
XGBoost	GHS04	safety_recall90	0.730721	28	0.9951	0.9841	0.9999	0.8092	0.7188	0.7241	0.6389	0.8214
XGBoost	GHS05	default_0.5	0.5	4333	0.8924	0.8868	0.8979	0.7373	0.6304	0.545	0.5501	0.7381
XGBoost	GHS05	calibrated_F1	0.637968	4333	0.8924	0.8868	0.8979	0.7373	0.6654	0.5956	0.6837	0.648
XGBoost	GHS05	safety_recall90	0.258556	4333	0.8924	0.8868	0.8979	0.7373	0.5009	0.4122	0.3452	0.9128
XGBoost	GHS06	default_0.5	0.5	1520	0.8296	0.818	0.8402	0.3262	0.3353	0.3144	0.2273	0.6388
XGBoost	GHS06	calibrated_F1	0.633534	1520	0.8296	0.818	0.8402	0.3262	0.3612	0.3162	0.3352	0.3914
XGBoost	GHS06	safety_recall90	0.30901	1520	0.8296	0.818	0.8402	0.3262	0.2049	0.21	0.116	0.8803
XGBoost	GHS07	default_0.5	0.5	21967	0.8336	0.8261	0.8412	0.9765	0.8611	0.3401	0.9638	0.7781
XGBoost	GHS07	calibrated_F1	0.111349	21967	0.8336	0.8261	0.8412	0.9765	0.9502	0.202	0.9083	0.9962
XGBoost	GHS07	safety_recall90	0.295102	21967	0.8336	0.8261	0.8412	0.9765	0.9281	0.3268	0.9371	0.9193
XGBoost	GHS08	default_0.5	0.5	1360	0.9004	0.8927	0.9081	0.4518	0.3806	0.3833	0.2558	0.7434
XGBoost	GHS08	calibrated_F1	0.782329	1360	0.9004	0.8927	0.9081	0.4518	0.4668	0.4341	0.4406	0.4963
XGBoost	GHS08	safety_recall90	0.270562	1360	0.9004	0.8927	0.9081	0.4518	0.2765	0.3097	0.1638	0.8875
XGBoost	GHS09	default_0.5	0.5	1773	0.8969	0.8903	0.9033	0.4586	0.4334	0.418	0.3039	0.7552
XGBoost	GHS09	calibrated_F1	0.792764	1773	0.8969	0.8903	0.9033	0.4586	0.474	0.4307	0.4359	0.5195
XGBoost	GHS09	safety_recall90	0.295801	1773	0.8969	0.8903	0.9033	0.4586	0.371	0.3775	0.2367	0.8573
SVM	GHS01	default_0.5	0.5	18	0.9822	0.959	0.9984	0.4446	0.3509	0.3768	0.2564	0.5556
SVM	GHS01	calibrated_F1	0.922207	18	0.9822	0.959	0.9984	0.4446	0.3636	0.4713	1.0	0.2222
SVM	GHS01	safety_recall90	0.000997	18	0.9822	0.959	0.9984	0.4446	0.0029	0.0271	0.0015	1.0
SVM	GHS02	default_0.5	0.5	1063	0.934	0.9254	0.9418	0.5797	0.5254	0.5236	0.6801	0.428
SVM	GHS02	calibrated_F1	0.230508	1063	0.934	0.9254	0.9418	0.5797	0.5528	0.5398	0.4626	0.6867
SVM	GHS02	safety_recall90	0.024989	1063	0.934	0.9254	0.9418	0.5797	0.3226	0.3696	0.1976	0.8777
SVM	GHS03	default_0.5	0.5	41	0.9295	0.8628	0.9844	0.3229	0.2646	0.319	0.1689	0.6098
SVM	GHS03	calibrated_F1	0.808422	41	0.9295	0.8628	0.9844	0.3229	0.3621	0.3773	0.28	0.5122
SVM	GHS03	safety_recall90	0.02138	41	0.9295	0.8628	0.9844	0.3229	0.0826	0.1857	0.0435	0.8293
SVM	GHS04	default_0.5	0.5	28	0.9908	0.9725	0.9993	0.5309	0.3898	0.4572	0.2556	0.8214
SVM	GHS04	calibrated_F1	0.875264	28	0.9908	0.9725	0.9993	0.5309	0.5763	0.5765	0.5484	0.6071
SVM	GHS04	safety_recall90	0.316053	28	0.9908	0.9725	0.9993	0.5309	0.3268	0.4215	0.2	0.8929
SVM	GHS05	default_0.5	0.5	4333	0.8551	0.8486	0.8619	0.6395	0.5762	0.5158	0.7041	0.4877
SVM	GHS05	calibrated_F1	0.276596	4333	0.8551	0.8486	0.8619	0.6395	0.5915	0.4963	0.5427	0.6499
SVM	GHS05	safety_recall90	0.058622	4333	0.8551	0.8486	0.8619	0.6395	0.4542	0.3489	0.303	0.9065

Model	Pictogram Code	Threshold Type	Threshold Value	N Test Positive	AUC ROC	AUC CI95 lower	AUC CI95 upper	Average Precision	F1	MCC	Precision	Recall
SVM	GHS06	default_0.5	0.5	1520	0.7425	0.729	0.7555	0.1868	0.0117	0.0575	0.6429	0.0059
SVM	GHS06	calibrated_F1	0.134883	1520	0.7425	0.729	0.7555	0.1868	0.264	0.21	0.2069	0.3645
SVM	GHS06	safety_recall90	0.026945	1520	0.7425	0.729	0.7555	0.1868	0.1595	0.1322	0.0879	0.8618
SVM	GHS07	default_0.5	0.5	21967	0.7681	0.7582	0.7778	0.9647	0.9481	0.0867	0.904	0.9967
SVM	GHS07	calibrated_F1	0.316579	21967	0.7681	0.7582	0.7778	0.9647	0.9485	0.0281	0.9022	1.0
SVM	GHS07	safety_recall90	0.740947	21967	0.7681	0.7582	0.7778	0.9647	0.927	0.2452	0.9257	0.9283
SVM	GHS08	default_0.5	0.5	1360	0.8062	0.7948	0.8181	0.2579	0.0634	0.1208	0.5111	0.0338
SVM	GHS08	calibrated_F1	0.153138	1360	0.8062	0.7948	0.8181	0.2579	0.3392	0.2977	0.2956	0.3978
SVM	GHS08	safety_recall90	0.022405	1360	0.8062	0.7948	0.8181	0.2579	0.1728	0.1791	0.0958	0.8831
SVM	GHS09	default_0.5	0.5	1773	0.8319	0.8224	0.8414	0.3068	0.1831	0.1994	0.4484	0.1151
SVM	GHS09	calibrated_F1	0.292824	1773	0.8319	0.8224	0.8414	0.3068	0.3688	0.3173	0.3533	0.3858
SVM	GHS09	safety_recall90	0.035971	1773	0.8319	0.8224	0.8414	0.3068	0.2834	0.2755	0.1717	0.8111
RandomForest_NoClassWeight	GHS01	default_0.5	0.5	18	0.9891	0.9815	0.9956	0.3194	0.0	0.0	0.0	0.0
RandomForest_NoClassWeight	GHS01	calibrated_F1	0.089656	18	0.9891	0.9815	0.9956	0.3194	0.4167	0.4809	0.8333	0.2778
RandomForest_NoClassWeight	GHS01	safety_recall90	0.000488	18	0.9891	0.9815	0.9956	0.3194	0.0068	0.0519	0.0034	1.0
RandomForest_NoClassWeight	GHS02	default_0.5	0.5	1063	0.9638	0.9585	0.9686	0.7094	0.5356	0.5664	0.8571	0.3895
RandomForest_NoClassWeight	GHS02	calibrated_F1	0.280639	1063	0.9638	0.9585	0.9686	0.7094	0.6591	0.6433	0.6393	0.6802
RandomForest_NoClassWeight	GHS02	safety_recall90	0.08159	1063	0.9638	0.9585	0.9686	0.7094	0.46	0.4953	0.3092	0.8984
RandomForest_NoClassWeight	GHS03	default_0.5	0.5	41	0.9282	0.8657	0.9834	0.443	0.3265	0.4414	1.0	0.1951
RandomForest_NoClassWeight	GHS03	calibrated_F1	0.223356	41	0.9282	0.8657	0.9834	0.443	0.4262	0.4534	0.65	0.3171
RandomForest_NoClassWeight	GHS03	safety_recall90	0.007893	41	0.9282	0.8657	0.9834	0.443	0.0943	0.1939	0.0502	0.7805
RandomForest_NoClassWeight	GHS04	default_0.5	0.5	28	0.9782	0.9284	0.9996	0.6607	0.0	0.0	0.0	0.0
RandomForest_NoClassWeight	GHS04	calibrated_F1	0.124921	28	0.9782	0.9284	0.9996	0.6607	0.5783	0.611	0.4364	0.8571
RandomForest_NoClassWeight	GHS04	safety_recall90	0.100491	28	0.9782	0.9284	0.9996	0.6607	0.5275	0.5708	0.381	0.8571
RandomForest_NoClassWeight	GHS05	default_0.5	0.5	4333	0.8823	0.8767	0.888	0.7264	0.5372	0.5389	0.895	0.3838
RandomForest_NoClassWeight	GHS05	calibrated_F1	0.284526	4333	0.8823	0.8767	0.888	0.7264	0.6504	0.5766	0.6633	0.6379

Model	Pictogram Code	Threshold Type	Threshold Value	N Test Positive	AUC ROC	AUC CI95 lower	AUC CI95 upper	Average Precision	F1	MCC	Precision	Recall
RandomForest_NoClassWeight	GHS05	safety_recall90	0.108706	4333	0.8823	0.8767	0.888	0.7264	0.4733	0.3763	0.3194	0.9128
RandomForest_NoClassWeight	GHS06	default_0.5	0.5	1520	0.8293	0.8181	0.8396	0.3164	0.0208	0.0994	1.0	0.0105
RandomForest_NoClassWeight	GHS06	calibrated_F1	0.166874	1520	0.8293	0.8181	0.8396	0.3164	0.337	0.2966	0.3603	0.3164
RandomForest_NoClassWeight	GHS06	safety_recall90	0.038515	1520	0.8293	0.8181	0.8396	0.3164	0.1999	0.2043	0.1126	0.8882
RandomForest_NoClassWeight	GHS07	default_0.5	0.5	21967	0.8284	0.8204	0.8366	0.9753	0.95	0.1645	0.9051	0.9995
RandomForest_NoClassWeight	GHS07	calibrated_F1	0.645515	21967	0.8284	0.8204	0.8366	0.9753	0.9504	0.24	0.9117	0.9927
RandomForest_NoClassWeight	GHS07	safety_recall90	0.79188	21967	0.8284	0.8204	0.8366	0.9753	0.9283	0.3402	0.9391	0.9178
RandomForest_NoClassWeight	GHS08	default_0.5	0.5	1360	0.8917	0.8832	0.9006	0.4464	0.165	0.2715	0.8671	0.0912
RandomForest_NoClassWeight	GHS08	calibrated_F1	0.258257	1360	0.8917	0.8832	0.9006	0.4464	0.4314	0.4099	0.5175	0.3699
RandomForest_NoClassWeight	GHS08	safety_recall90	0.035427	1360	0.8917	0.8832	0.9006	0.4464	0.2401	0.2721	0.1385	0.9022
RandomForest_NoClassWeight	GHS09	default_0.5	0.5	1773	0.8794	0.8718	0.8873	0.4496	0.1399	0.2249	0.7405	0.0773
RandomForest_NoClassWeight	GHS09	calibrated_F1	0.231742	1773	0.8794	0.8718	0.8873	0.4496	0.4638	0.4196	0.4273	0.5071
RandomForest_NoClassWeight	GHS09	safety_recall90	0.081373	1773	0.8794	0.8718	0.8873	0.4496	0.3642	0.364	0.2338	0.824

This table has 22 columns in full; 13 are shown here and 9 are omitted so the table stays legible. The complete table, with every column, is sheet 'S3_full_performance' of the accompanying Excel workbook.

108 rows. Source sheet: S3_full_performance.

Table S4

The twenty most influential SHAP features per hazard class.

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP	Description
GHS01_Explosive	1	MACCS_70	0.55366	-0.463924	MACCS substructure key 70: matches SMARTS pattern <chem>[!#6;!#1]~[#7]~[!#6;!#1]</chem>
GHS01_Explosive	2	MACCS_124	0.490891	-0.323515	MACCS substructure key 124: matches SMARTS pattern <chem>[!#6;!#1]~[!#6;!#1]</chem>
GHS01_Explosive	3	TPSA	0.275038	-0.269869	topological polar surface area - the size of the polar region
GHS01_Explosive	4	MACCS_142	0.166226	-0.160253	MACCS substructure key 142: matches SMARTS pattern <chem>[#7]</chem>
GHS01_Explosive	5	MACCS_102	0.140569	-0.039449	MACCS substructure key 102: matches SMARTS pattern <chem>[!#6;!#1]~[#8]</chem>
GHS01_Explosive	6	Chi2n	0.075281	-0.061007	second-order connectivity index
GHS01_Explosive	7	Kappa3	0.070183	-0.062226	third-order shape index - centrality of branching
GHS01_Explosive	8	MACCS_49	0.067073	-0.058121	MACCS substructure key 49: matches SMARTS pattern <chem>[+0]</chem>
GHS01_Explosive	9	MACCS_148	0.060744	-0.005539	MACCS substructure key 148: matches SMARTS pattern <chem>*~[!#6;!#1](-~)*</chem>
GHS01_Explosive	10	MACCS_83	0.055498	-0.024283	MACCS substructure key 83: matches SMARTS pattern <chem>[!#6;!#1]1~*~*~*~*~1</chem>
GHS01_Explosive	11	MACCS_90	0.045773	-0.044074	MACCS substructure key 90: matches SMARTS pattern <chem>\$([!#6;!#1;!H0]~*~*~[CH2]~*),\$([!#6;!#1;!H0;R]1@[R]@[R]@[CH2;R]1),\$([!#6;!#1;!H0]~[R]1@[R]@[CH2;R]1)\$</chem>
GHS01_Explosive	12	MACCS_156	0.035559	-0.020667	MACCS substructure key 156: matches SMARTS pattern <chem>[#7]~*(~*)~*</chem>
GHS01_Explosive	13	Morgan_514	0.035028	-0.034336	ECFP4 circular substructure bit 514 (radius 2 atom environment)
GHS01_Explosive	14	MACCS_154	0.034543	-0.025669	MACCS substructure key 154: matches SMARTS pattern <chem>[#6]=[#8]</chem>
GHS01_Explosive	15	Morgan_128	0.030807	-0.01278	ECFP4 circular substructure bit 128 (radius 2 atom environment)
GHS01_Explosive	16	MACCS_117	0.03029	-0.013108	MACCS substructure key 117: matches SMARTS pattern <chem>[#7]~*~[#8]</chem>
GHS01_Explosive	17	MACCS_111	0.02967	-0.020643	MACCS substructure key 111: matches SMARTS pattern <chem>[#7]~*~[CH2]~*</chem>
GHS01_Explosive	18	MACCS_121	0.029436	-0.021602	MACCS substructure key 121: matches SMARTS pattern <chem>[#7;R]</chem>
GHS01_Explosive	19	MACCS_161	0.028044	-0.027391	MACCS substructure key 161: matches SMARTS pattern <chem>[#7]</chem>

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP	Description
GHS01_Explosive	20	NumHAcceptors	0.027608	-0.025232	number of hydrogen-bond acceptors (O, N lone pairs)
GHS02_Flammable	1	BertzCT	0.649036	-0.375776	structural complexity
GHS02_Flammable	2	TPSA	0.540827	-0.349749	topological polar surface area - the size of the polar region
GHS02_Flammable	3	LabuteASA	0.439609	-0.25814	approximate molecular surface area
GHS02_Flammable	4	NumHDonors	0.233505	-0.07345	number of hydrogen-bond donors (OH, NH groups)
GHS02_Flammable	5	NHOHCount	0.181492	-0.073873	number of NH and OH groups
GHS02_Flammable	6	MolWt	0.172027	-0.122351	average molecular mass
GHS02_Flammable	7	MACCS_131	0.169805	-0.109887	MACCS substructure key 131: matches SMARTS pattern [#6;!#1;!H0]
GHS02_Flammable	8	ExactMolWt	0.161357	-0.122906	monoisotopic molecular mass
GHS02_Flammable	9	Morgan_650	0.152775	-0.052726	ECFP4 circular substructure bit 650 (radius 2 atom environment)
GHS02_Flammable	10	FractionCSP3	0.152108	-0.069067	fraction of carbons that are sp3 - three-dimensionality
GHS02_Flammable	11	Morgan_356	0.138924	-0.059179	ECFP4 circular substructure bit 356 (radius 2 atom environment)
GHS02_Flammable	12	MACCS_103	0.120147	-0.028014	MACCS substructure key 103: matches SMARTS pattern Cl
GHS02_Flammable	13	Chi4n	0.117017	-0.087387	fourth-order connectivity index
GHS02_Flammable	14	NumHeteroatoms	0.090413	-0.023964	number of atoms that are neither carbon nor hydrogen
GHS02_Flammable	15	MACCS_154	0.087076	-0.030397	MACCS substructure key 154: matches SMARTS pattern [#6]=[#8]
GHS02_Flammable	16	MACCS_88	0.087028	0.000504	MACCS substructure key 88: matches SMARTS pattern [#16]
GHS02_Flammable	17	RingCount	0.084934	-0.024075	total number of rings
GHS02_Flammable	18	MACCS_124	0.083471	-0.022454	MACCS substructure key 124: matches SMARTS pattern [#6;!#1]-[#6;!#1]
GHS02_Flammable	19	MACCS_149	0.081658	-0.008876	MACCS substructure key 149: matches SMARTS pattern [C;H3,H4]
GHS02_Flammable	20	Chi3n	0.080891	-0.054113	third-order connectivity index
GHS03_Oxidising	1	Chi3n	0.952689	-0.900104	third-order connectivity index
GHS03_Oxidising	2	MACCS_102	0.77605	-0.539722	MACCS substructure key 102: matches SMARTS pattern [#6;!#1]-[#8]
GHS03_Oxidising	3	TPSA	0.611993	-0.58199	topological polar surface area - the size of the polar region
GHS03_Oxidising	4	MACCS_124	0.455176	-0.396334	MACCS substructure key 124: matches SMARTS pattern [#6;!#1]-[#6;!#1]
GHS03_Oxidising	5	Kappa3	0.446737	-0.442095	third-order shape index - centrality of branching

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP	Description
GHS03_Oxidising	6	FractionCSP3	0.405366	-0.404242	fraction of carbons that are sp3 - three-dimensionality
GHS03_Oxidising	7	MACCS_31	0.375778	-0.310106	MACCS substructure key 31: matches SMARTS pattern [#6;!#1]-[F,Cl,Br,I]
GHS03_Oxidising	8	MolLogP	0.373483	-0.373384	lipophilicity - how strongly the compound prefers oil to water
GHS03_Oxidising	9	BalabanJ	0.365839	-0.365465	topological connectivity index - branching pattern
GHS03_Oxidising	10	Chi4n	0.350112	-0.331174	fourth-order connectivity index
GHS03_Oxidising	11	Kappa2	0.296654	-0.294025	second-order shape index - how linear or star-like the skeleton is
GHS03_Oxidising	12	BertzCT	0.281685	-0.281145	structural complexity
GHS03_Oxidising	13	MACCS_49	0.2722	-0.182807	MACCS substructure key 49: matches SMARTS pattern [!+0]
GHS03_Oxidising	14	MACCS_146	0.261383	-0.1793	MACCS substructure key 146: matches SMARTS pattern [#8]
GHS03_Oxidising	15	MACCS_88	0.234421	-0.084734	MACCS substructure key 88: matches SMARTS pattern [#16]
GHS03_Oxidising	16	LabuteASA	0.229907	-0.229907	approximate molecular surface area
GHS03_Oxidising	17	Kappa1	0.227362	-0.226283	first-order shape index - overall size
GHS03_Oxidising	18	Morgan_378	0.221343	-0.145816	ECFP4 circular substructure bit 378 (radius 2 atom environment)
GHS03_Oxidising	19	MACCS_130	0.214334	-0.14282	MACCS substructure key 130: matches SMARTS pattern [#6;!#1]-[!#6;!#1]
GHS03_Oxidising	20	MolWt	0.203915	-0.203915	average molecular mass
GHS04_CompressedGas	1	Kappa2	1.79048	-1.78889	second-order shape index - how linear or star-like the skeleton is
GHS04_CompressedGas	2	Chi4n	1.539784	-1.39107	fourth-order connectivity index
GHS04_CompressedGas	3	LabuteASA	1.355818	-1.354701	approximate molecular surface area
GHS04_CompressedGas	4	MolWt	0.767101	-0.766134	average molecular mass
GHS04_CompressedGas	5	ExactMolWt	0.74815	-0.746992	monoisotopic molecular mass
GHS04_CompressedGas	6	Chi2n	0.684677	-0.682578	second-order connectivity index
GHS04_CompressedGas	7	TPSA	0.661865	-0.649623	topological polar surface area - the size of the polar region
GHS04_CompressedGas	8	BertzCT	0.62329	-0.621016	structural complexity
GHS04_CompressedGas	9	Chi3n	0.53796	-0.53156	third-order connectivity index
GHS04_CompressedGas	10	MolLogP	0.520711	-0.520711	lipophilicity - how strongly the compound prefers oil to water
GHS04_CompressedGas	11	NumRotatableBonds	0.467301	-0.438768	molecular flexibility
GHS04_CompressedGas	12	Kappa3	0.448049	-0.438488	third-order shape index - centrality of branching

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP	Description
GHS04_CompressedGas	13	NumHDonors	0.417581	-0.399102	number of hydrogen-bond donors (OH, NH groups)
GHS04_CompressedGas	14	NumHeteroatoms	0.382897	-0.303319	number of atoms that are neither carbon nor hydrogen
GHS04_CompressedGas	15	BalabanJ	0.375097	-0.371112	topological connectivity index - branching pattern
GHS04_CompressedGas	16	Morgan_356	0.294656	-0.287391	ECFP4 circular substructure bit 356 (radius 2 atom environment)
GHS04_CompressedGas	17	NumHAcceptors	0.292755	-0.272068	number of hydrogen-bond acceptors (O, N lone pairs)
GHS04_CompressedGas	18	Morgan_33	0.292084	-0.231596	ECFP4 circular substructure bit 33 (radius 2 atom environment)
GHS04_CompressedGas	19	Kappa1	0.291974	0.100052	first-order shape index - overall size
GHS04_CompressedGas	20	NumAromaticRings	0.239348	-0.236859	number of aromatic rings
GHS05_Corrosive	1	BertzCT	0.345213	-0.102352	structural complexity
GHS05_Corrosive	2	Morgan_514	0.235701	-0.120401	ECFP4 circular substructure bit 514 (radius 2 atom environment)
GHS05_Corrosive	3	MACCS_31	0.153432	-0.027762	MACCS substructure key 31: matches SMARTS pattern <chem>[!#6;!#1]-[F,Cl,Br,I]</chem>
GHS05_Corrosive	4	MACCS_154	0.12328	0.000308	MACCS substructure key 154: matches SMARTS pattern <chem>[#6]=[#8]</chem>
GHS05_Corrosive	5	MACCS_153	0.113646	-0.037132	MACCS substructure key 153: matches SMARTS pattern <chem>[!#6;!#1]-[CH2]-*</chem>
GHS05_Corrosive	6	MACCS_117	0.096806	-0.00408	MACCS substructure key 117: matches SMARTS pattern <chem>[#7]-*-[#8]</chem>
GHS05_Corrosive	7	RingCount	0.092477	-0.023207	total number of rings
GHS05_Corrosive	8	MACCS_139	0.091429	-0.010479	MACCS substructure key 139: matches SMARTS pattern <chem>[O;!H0]</chem>
GHS05_Corrosive	9	BalabanJ	0.086371	-0.030852	topological connectivity index - branching pattern
GHS05_Corrosive	10	TPSA	0.084789	-0.007402	topological polar surface area - the size of the polar region
GHS05_Corrosive	11	MACCS_100	0.079445	-0.004686	MACCS substructure key 100: matches SMARTS pattern <chem>*-[CH2]-[#7]</chem>
GHS05_Corrosive	12	Morgan_659	0.070868	-0.034967	ECFP4 circular substructure bit 659 (radius 2 atom environment)
GHS05_Corrosive	13	MACCS_96	0.068252	-0.020786	MACCS substructure key 96: matches SMARTS pattern <chem>*1~*~*~*~*~1</chem>
GHS05_Corrosive	14	MACCS_151	0.067597	0.004427	MACCS substructure key 151: matches SMARTS pattern <chem>[#7;!H0]</chem>
GHS05_Corrosive	15	FractionCSP3	0.067205	-0.009668	fraction of carbons that are sp3 - three-dimensionality
GHS05_Corrosive	16	MACCS_158	0.066705	-0.001137	MACCS substructure key 158: matches SMARTS pattern <chem>[#6]-[#7]</chem>

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP	Description
GHS05_Corrosive	17	Morgan_85	0.065138	-0.011784	ECFP4 circular substructure bit 85 (radius 2 atom environment)
GHS05_Corrosive	18	MACCS_84	0.062641	-0.008013	MACCS substructure key 84: matches SMARTS pattern [NH2]
GHS05_Corrosive	19	Morgan_251	0.061808	-0.006315	ECFP4 circular substructure bit 251 (radius 2 atom environment)
GHS05_Corrosive	20	Morgan_356	0.06107	0.001274	ECFP4 circular substructure bit 356 (radius 2 atom environment)
GHS06_AcuteToxicity	1	ExactMolWt	0.171272	-0.080114	monoisotopic molecular mass
GHS06_AcuteToxicity	2	MACCS_41	0.090883	-0.039571	MACCS substructure key 41: matches SMARTS pattern [#6]#[#7]
GHS06_AcuteToxicity	3	MACCS_164	0.077375	-0.012709	MACCS substructure key 164: matches SMARTS pattern [#8]
GHS06_AcuteToxicity	4	Chi3n	0.076763	-0.020702	third-order connectivity index
GHS06_AcuteToxicity	5	Chi2n	0.067517	-0.018624	second-order connectivity index
GHS06_AcuteToxicity	6	BalabanJ	0.052652	-0.023093	topological connectivity index - branching pattern
GHS06_AcuteToxicity	7	MACCS_119	0.051779	-0.008637	MACCS substructure key 119: matches SMARTS pattern [#7]=*
GHS06_AcuteToxicity	8	MACCS_123	0.050552	-0.015656	MACCS substructure key 123: matches SMARTS pattern [#8]~[#6]~[#8]
GHS06_AcuteToxicity	9	MACCS_137	0.047352	-0.030456	MACCS substructure key 137: matches SMARTS pattern [!C;!c;R]
GHS06_AcuteToxicity	10	Chi1	0.044922	-0.009751	first-order connectivity index - bond-level branching
GHS06_AcuteToxicity	11	Morgan_926	0.040315	-0.010392	ECFP4 circular substructure bit 926 (radius 2 atom environment)
GHS06_AcuteToxicity	12	MACCS_157	0.035443	-0.008175	MACCS substructure key 157: matches SMARTS pattern [#6]-[#8]
GHS06_AcuteToxicity	13	MolWt	0.032215	-0.020934	average molecular mass
GHS06_AcuteToxicity	14	Chi4n	0.031492	-0.016013	fourth-order connectivity index
GHS06_AcuteToxicity	15	BertzCT	0.029485	-0.022315	structural complexity
GHS06_AcuteToxicity	16	Chi0	0.027305	-0.001726	zeroth-order connectivity index - atom count weighted by degree
GHS06_AcuteToxicity	17	TPSA	0.026778	-0.009967	topological polar surface area - the size of the polar region
GHS06_AcuteToxicity	18	MACCS_120	0.026268	-0.010285	MACCS substructure key 120: matches SMARTS pattern [!#6;R]
GHS06_AcuteToxicity	19	NHOHCount	0.025909	-0.007294	number of NH and OH groups
GHS06_AcuteToxicity	20	FractionCSP3	0.02488	-0.012069	fraction of carbons that are sp3 - three-dimensionality
GHS07_Irritant	1	Chi1	0.164398	0.061072	first-order connectivity index - bond-level branching

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP	Description
GHS07_Irritant	2	MACCS_137	0.153679	0.058604	MACCS substructure key 137: matches SMARTS pattern [!C;!c;R]
GHS07_Irritant	3	LabuteASA	0.13757	0.047797	approximate molecular surface area
GHS07_Irritant	4	BertzCT	0.11732	0.038474	structural complexity
GHS07_Irritant	5	MolLogP	0.115468	0.020777	lipophilicity - how strongly the compound prefers oil to water
GHS07_Irritant	6	RingCount	0.105317	0.034953	total number of rings
GHS07_Irritant	7	Morgan_514	0.098046	0.037188	ECFP4 circular substructure bit 514 (radius 2 atom environment)
GHS07_Irritant	8	MACCS_121	0.097865	0.018133	MACCS substructure key 121: matches SMARTS pattern [#7;R]
GHS07_Irritant	9	NumHDonors	0.090397	0.02289	number of hydrogen-bond donors (OH, NH groups)
GHS07_Irritant	10	MACCS_65	0.077758	0.019894	MACCS substructure key 65: matches SMARTS pattern c:n
GHS07_Irritant	11	MACCS_31	0.060578	0.021937	MACCS substructure key 31: matches SMARTS pattern [!#6;!#1]~[F,Cl,Br,I]
GHS07_Irritant	12	MACCS_154	0.058368	0.013593	MACCS substructure key 154: matches SMARTS pattern [#6]=[#8]
GHS07_Irritant	13	MolWt	0.056935	0.027487	average molecular mass
GHS07_Irritant	14	MACCS_139	0.052449	0.002712	MACCS substructure key 139: matches SMARTS pattern [O;!H0]
GHS07_Irritant	15	MACCS_123	0.048644	0.017686	MACCS substructure key 123: matches SMARTS pattern [#8]~[#6]~[#8]
GHS07_Irritant	16	NumHeteroatoms	0.046305	0.018333	number of atoms that are neither carbon nor hydrogen
GHS07_Irritant	17	ExactMolWt	0.044717	0.023948	monoisotopic molecular mass
GHS07_Irritant	18	Chi4n	0.043983	0.021953	fourth-order connectivity index
GHS07_Irritant	19	Chi3n	0.042163	0.02033	third-order connectivity index
GHS07_Irritant	20	TPSA	0.042063	0.015136	topological polar surface area - the size of the polar region
GHS08_HealthHazard	1	Chi4n	0.175769	-0.100279	fourth-order connectivity index
GHS08_HealthHazard	2	TPSA	0.140887	-0.071943	topological polar surface area - the size of the polar region
GHS08_HealthHazard	3	NumHDonors	0.121378	-0.03439	number of hydrogen-bond donors (OH, NH groups)
GHS08_HealthHazard	4	MACCS_121	0.118889	-0.027694	MACCS substructure key 121: matches SMARTS pattern [#7;R]
GHS08_HealthHazard	5	MACCS_119	0.115361	-0.065893	MACCS substructure key 119: matches SMARTS pattern [#7]=*
GHS08_HealthHazard	6	Chi1	0.106844	-0.037051	first-order connectivity index - bond-level branching
GHS08_HealthHazard	7	Chi2n	0.09944	-0.053133	second-order connectivity index

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP	Description
GHS08_HealthHazard	8	Morgan_893	0.099022	-0.020896	ECFP4 circular substructure bit 893 (radius 2 atom environment)
GHS08_HealthHazard	9	Morgan_725	0.094818	-0.035108	ECFP4 circular substructure bit 725 (radius 2 atom environment)
GHS08_HealthHazard	10	Chi3n	0.083487	-0.056616	third-order connectivity index
GHS08_HealthHazard	11	MACCS_44	0.079207	-0.038349	MACCS substructure key 44: matches SMARTS pattern [#1;!#6;!#7;!#8;!#9;!#14;!#15;!#16;!#17;!#35;!#53]
GHS08_HealthHazard	12	Chi0	0.076798	-0.037701	zeroth-order connectivity index - atom count weighted by degree
GHS08_HealthHazard	13	NumHeteroatoms	0.0735	-0.041089	number of atoms that are neither carbon nor hydrogen
GHS08_HealthHazard	14	RingCount	0.071652	-0.02386	total number of rings
GHS08_HealthHazard	15	Kappa1	0.071568	-0.029936	first-order shape index - overall size
GHS08_HealthHazard	16	MACCS_134	0.069779	-0.003579	MACCS substructure key 134: matches SMARTS pattern [F,Cl,Br,I]
GHS08_HealthHazard	17	ExactMolWt	0.064652	-0.031146	monoisotopic molecular mass
GHS08_HealthHazard	18	LabuteASA	0.061388	-0.019984	approximate molecular surface area
GHS08_HealthHazard	19	MolWt	0.060529	-0.036814	average molecular mass
GHS08_HealthHazard	20	FractionCSP3	0.058271	-0.030057	fraction of carbons that are sp3 - three-dimensionality
GHS09_Environmental	1	MolLogP	0.378431	-0.133164	lipophilicity - how strongly the compound prefers oil to water
GHS09_Environmental	2	MACCS_137	0.358769	-0.145773	MACCS substructure key 137: matches SMARTS pattern [!C;!c;R]
GHS09_Environmental	3	Chi1	0.30195	-0.145785	first-order connectivity index - bond-level branching
GHS09_Environmental	4	LabuteASA	0.25488	-0.145089	approximate molecular surface area
GHS09_Environmental	5	MACCS_65	0.129309	-0.055423	MACCS substructure key 65: matches SMARTS pattern c:n
GHS09_Environmental	6	NumAromaticRings	0.128962	-0.031911	number of aromatic rings
GHS09_Environmental	7	Morgan_514	0.12152	-0.055874	ECFP4 circular substructure bit 514 (radius 2 atom environment)
GHS09_Environmental	8	MACCS_44	0.112232	-0.040265	MACCS substructure key 44: matches SMARTS pattern [#1;!#6;!#7;!#8;!#9;!#14;!#15;!#16;!#17;!#35;!#53]
GHS09_Environmental	9	MACCS_18	0.103174	-0.072472	MACCS substructure key 18: matches SMARTS pattern [#5,#13,#31,#49,#81]
GHS09_Environmental	10	FractionCSP3	0.102216	-0.039803	fraction of carbons that are sp3 - three-dimensionality
GHS09_Environmental	11	Kappa2	0.098929	-0.054517	second-order shape index - how linear or star-like the skeleton is

GHS Column	Rank	Feature	Mean Abs SHAP	Mean Signed SHAP	Description
GHS09_Environmental	12	MACCS_46	0.085896	-0.015112	MACCS substructure key 46: matches SMARTS pattern Br
GHS09_Environmental	13	Morgan_378	0.078924	-0.00221	ECFP4 circular substructure bit 378 (radius 2 atom environment)
GHS09_Environmental	14	TPSA	0.071746	-0.017135	topological polar surface area - the size of the polar region
GHS09_Environmental	15	Morgan_389	0.07161	0.000231	ECFP4 circular substructure bit 389 (radius 2 atom environment)
GHS09_Environmental	16	MACCS_134	0.068983	-0.004196	MACCS substructure key 134: matches SMARTS pattern [F,Cl,Br,I]
GHS09_Environmental	17	MACCS_145	0.066914	-0.033925	MACCS substructure key 145: matches SMARTS pattern *1~*~*~*~*~1
GHS09_Environmental	18	NumHeteroatoms	0.0636	-0.018185	number of atoms that are neither carbon nor hydrogen
GHS09_Environmental	19	Chi0	0.058698	-0.017893	zeroth-order connectivity index - atom count weighted by degree
GHS09_Environmental	20	Chi4n	0.058114	-0.02602	fourth-order connectivity index

This table has 9 columns in full; 6 are shown here and 3 are omitted so the table stays legible. The complete table, with every column, is sheet 'S4_SHAP_top20' of the accompanying Excel workbook. 180 rows. Source sheet: S4_SHAP_top20.

Table S4b

Chemical interpretation of the five most influential features per hazard class.

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS01	1	MACCS_70	0.55366	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	MACCS substructure key 70: matches SMARTS pattern <chem>[!#6;!#1]~[#7]~[!#6;!#1]</chem>	Explosivity comes from functional groups that store chemical energy and can release it without external oxygen - nitro, nitrate ester, azide, peroxide and strained-ring systems. A high nitrogen and oxygen count relative to carbon is the classic signature.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS01	2	MACCS_124	0.490891	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	MACCS substructure key 124: matches SMARTS pattern <chem>[!#6;!#1]~[!#6;!#1]</chem>	Explosivity comes from functional groups that store chemical energy and can release it without external oxygen - nitro, nitrate ester, azide, peroxide and strained-ring systems. A high nitrogen and oxygen count relative to carbon is the classic signature.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS01	3	TPSA	0.275038	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	topological polar surface area - the size of the polar region	Explosivity comes from functional groups that store chemical energy and can release it without external oxygen - nitro, nitrate ester, azide, peroxide and strained-ring systems. A high nitrogen and oxygen count relative to carbon is the classic signature.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS01	4	MACCS_142	0.166226	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	MACCS substructure key 142: matches SMARTS pattern <chem>[#7]</chem>	Explosivity comes from functional groups that store chemical energy and can release it without external oxygen - nitro, nitrate ester, azide, peroxide and strained-ring systems. A high nitrogen and oxygen count relative to carbon is the classic signature.	not among the classically expected bulk descriptors; likely acting as a structural marker

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS01	5	MACCS_102	0.140569	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	MACCS substructure key 102: matches SMARTS pattern <chem>[!#6;!#1]-[#8]</chem>	Explosivity comes from functional groups that store chemical energy and can release it without external oxygen - nitro, nitrate ester, azide, peroxide and strained-ring systems. A high nitrogen and oxygen count relative to carbon is the classic signature.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS02	1	BertzCT	0.649036	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	structural complexity	Flammability tracks volatility and hydrocarbon content. Small, light, lipophilic molecules with few polar groups have low flash points; high polarity and hydrogen bonding raise the boiling point and reduce flammability.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS02	2	TPSA	0.540827	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	topological polar surface area - the size of the polar region	Flammability tracks volatility and hydrocarbon content. Small, light, lipophilic molecules with few polar groups have low flash points; high polarity and hydrogen bonding raise the boiling point and reduce flammability.	yes - this is a descriptor a chemist would expect to matter for this hazard
GHS02	3	LabuteASA	0.439609	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	approximate molecular surface area	Flammability tracks volatility and hydrocarbon content. Small, light, lipophilic molecules with few polar groups have low flash points; high polarity and hydrogen bonding raise the boiling point and reduce flammability.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS02	4	NumHDonors	0.233505	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	number of hydrogen-bond donors (OH, NH groups)	Flammability tracks volatility and hydrocarbon content. Small, light, lipophilic molecules with few polar groups have low flash points; high polarity and hydrogen bonding raise the boiling point and reduce flammability.	not among the classically expected bulk descriptors; likely acting as a structural marker

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS02	5	NHOHCount	0.181492	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	number of NH and OH groups	Flammability tracks volatility and hydrocarbon content. Small, light, lipophilic molecules with few polar groups have low flash points; high polarity and hydrogen bonding raise the boiling point and reduce flammability.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS03	1	Chi3n	0.952689	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	third-order connectivity index	Oxidisers carry oxygen in a high formal oxidation state - peroxides, perchlorates, nitrates, chlorates, permanganates. A high oxygen-to-carbon ratio combined with low carbon content is the dominant signal.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS03	2	MACCS_102	0.77605	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	MACCS substructure key 102: matches SMARTS pattern <code>[!#6;!#1]~[#8]</code>	Oxidisers carry oxygen in a high formal oxidation state - peroxides, perchlorates, nitrates, chlorates, permanganates. A high oxygen-to-carbon ratio combined with low carbon content is the dominant signal.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS03	3	TPSA	0.611993	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	topological polar surface area - the size of the polar region	Oxidisers carry oxygen in a high formal oxidation state - peroxides, perchlorates, nitrates, chlorates, permanganates. A high oxygen-to-carbon ratio combined with low carbon content is the dominant signal.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS03	4	MACCS_124	0.455176	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	MACCS substructure key 124: matches SMARTS pattern <code>[!#6;!#1]~[!#6;!#1]</code>	Oxidisers carry oxygen in a high formal oxidation state - peroxides, perchlorates, nitrates, chlorates, permanganates. A high oxygen-to-carbon ratio combined with low carbon content is the dominant signal.	not among the classically expected bulk descriptors; likely acting as a structural marker

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS03	5	Kappa3	0.446737	non-monotonic - the descriptor matters, but its effect is not a simple increase or decrease	third-order shape index - centrality of branching	Oxidisers carry oxygen in a high formal oxidation state - peroxides, perchlorates, nitrates, chlorates, permanganates. A high oxygen-to-carbon ratio combined with low carbon content is the dominant signal.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS04	1	Kappa2	1.79048	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	second-order shape index - how linear or star-like the skeleton is	The compressed-gas pictogram is assigned on physical state rather than reactivity, so it is predicted almost entirely from molecular size. Very low molecular weight and very few heavy atoms are what a gas looks like to a descriptor calculation.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS04	2	Chi4n	1.539784	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	fourth-order connectivity index	The compressed-gas pictogram is assigned on physical state rather than reactivity, so it is predicted almost entirely from molecular size. Very low molecular weight and very few heavy atoms are what a gas looks like to a descriptor calculation.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS04	3	LabuteASA	1.355818	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	approximate molecular surface area	The compressed-gas pictogram is assigned on physical state rather than reactivity, so it is predicted almost entirely from molecular size. Very low molecular weight and very few heavy atoms are what a gas looks like to a descriptor calculation.	yes - this is a descriptor a chemist would expect to matter for this hazard
GHS04	4	MolWt	0.767101	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	average molecular mass	The compressed-gas pictogram is assigned on physical state rather than reactivity, so it is predicted almost entirely from molecular size. Very low molecular weight and very few heavy atoms are what a gas looks like to a descriptor calculation.	yes - this is a descriptor a chemist would expect to matter for this hazard

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS04	5	ExactMolWt	0.74815	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	monoisotopic molecular mass	The compressed-gas pictogram is assigned on physical state rather than reactivity, so it is predicted almost entirely from molecular size. Very low molecular weight and very few heavy atoms are what a gas looks like to a descriptor calculation.	yes - this is a descriptor a chemist would expect to matter for this hazard
GHS05	1	BertzCT	0.345213	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	structural complexity	Corrosivity is driven by extremes of pH and by strongly electrophilic groups: mineral acids, acid halides, anhydrides, quaternary hydroxides and amines. Ionisable groups and hydrogen-bond donors dominate.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS05	2	Morgan_514	0.235701	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	ECFP4 circular substructure bit 514; an example environment that sets it is Cl	Corrosivity is driven by extremes of pH and by strongly electrophilic groups: mineral acids, acid halides, anhydrides, quaternary hydroxides and amines. Ionisable groups and hydrogen-bond donors dominate.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS05	3	MACCS_31	0.153432	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	MACCS substructure key 31: matches SMARTS pattern [<i>!#6;!#1</i>]-[F,Cl,Br,I]	Corrosivity is driven by extremes of pH and by strongly electrophilic groups: mineral acids, acid halides, anhydrides, quaternary hydroxides and amines. Ionisable groups and hydrogen-bond donors dominate.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS05	4	MACCS_154	0.12328	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	MACCS substructure key 154: matches SMARTS pattern [<i>#6</i>]=[<i>#8</i>]	Corrosivity is driven by extremes of pH and by strongly electrophilic groups: mineral acids, acid halides, anhydrides, quaternary hydroxides and amines. Ionisable groups and hydrogen-bond donors dominate.	not among the classically expected bulk descriptors; likely acting as a structural marker

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS05	5	MACCS_153	0.113646	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	MACCS substructure key 153: matches SMARTS pattern <chem>[!#6;!#1]-[CH2]~*</chem>	Corrosivity is driven by extremes of pH and by strongly electrophilic groups: mineral acids, acid halides, anhydrides, quaternary hydroxides and amines. Ionisable groups and hydrogen-bond donors dominate.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS06	1	ExactMolWt	0.171272	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	monoisotopic molecular mass	Acute toxicity requires a molecule to reach a biological target quickly. Moderate lipophilicity aids membrane crossing, while specific toxicophores - organophosphates, cyanides, heavy-metal centres, alkaloid scaffolds - drive potency. Fingerprint bits usually outrank bulk properties for this class.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS06	2	MACCS_41	0.090883	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	MACCS substructure key 41: matches SMARTS pattern <chem>[#6]#[#7]</chem>	Acute toxicity requires a molecule to reach a biological target quickly. Moderate lipophilicity aids membrane crossing, while specific toxicophores - organophosphates, cyanides, heavy-metal centres, alkaloid scaffolds - drive potency. Fingerprint bits usually outrank bulk properties for this class.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS06	3	MACCS_164	0.077375	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	MACCS substructure key 164: matches SMARTS pattern <chem>[#8]</chem>	Acute toxicity requires a molecule to reach a biological target quickly. Moderate lipophilicity aids membrane crossing, while specific toxicophores - organophosphates, cyanides, heavy-metal centres, alkaloid scaffolds - drive potency. Fingerprint bits usually outrank bulk properties for this class.	not among the classically expected bulk descriptors; likely acting as a structural marker

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS06	4	Chi3n	0.076763	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	third-order connectivity index	Acute toxicity requires a molecule to reach a biological target quickly. Moderate lipophilicity aids membrane crossing, while specific toxicophores - organophosphates, cyanides, heavy-metal centres, alkaloid scaffolds - drive potency. Fingerprint bits usually outrank bulk properties for this class.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS06	5	Chi2n	0.067517	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	second-order connectivity index	Acute toxicity requires a molecule to reach a biological target quickly. Moderate lipophilicity aids membrane crossing, while specific toxicophores - organophosphates, cyanides, heavy-metal centres, alkaloid scaffolds - drive potency. Fingerprint bits usually outrank bulk properties for this class.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS07	1	Chi1	0.164398	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	first-order connectivity index - bond-level branching	The exclamation-mark pictogram covers skin, eye and respiratory irritation and lower-grade acute toxicity. It is the commonest and most chemically diverse class, so the model relies on a broad mix of polarity, reactivity and substructure features rather than any single driver.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS07	2	MACCS_137	0.153679	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	MACCS substructure key 137: matches SMARTS pattern [!C;!c;R]	The exclamation-mark pictogram covers skin, eye and respiratory irritation and lower-grade acute toxicity. It is the commonest and most chemically diverse class, so the model relies on a broad mix of polarity, reactivity and substructure features rather than any single driver.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS07	3	LabuteASA	0.13757	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	approximate molecular surface area	The exclamation-mark pictogram covers skin, eye and respiratory irritation and lower-grade acute toxicity. It is the commonest and most chemically diverse class, so the model relies on a broad mix of polarity, reactivity and substructure features rather than any single driver.	not among the classically expected bulk descriptors; likely acting as a structural marker

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS07	4	BertzCT	0.11732	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	structural complexity	The exclamation-mark pictogram covers skin, eye and respiratory irritation and lower-grade acute toxicity. It is the commonest and most chemically diverse class, so the model relies on a broad mix of polarity, reactivity and substructure features rather than any single driver.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS07	5	MolLogP	0.115468	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	lipophilicity - how strongly the compound prefers oil to water	The exclamation-mark pictogram covers skin, eye and respiratory irritation and lower-grade acute toxicity. It is the commonest and most chemically diverse class, so the model relies on a broad mix of polarity, reactivity and substructure features rather than any single driver.	yes - this is a descriptor a chemist would expect to matter for this hazard
GHS08	1	Chi4n	0.175769	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	fourth-order connectivity index	The health-hazard pictogram flags carcinogenicity, mutagenicity, reproductive toxicity, respiratory sensitisation and aspiration hazard. Planar aromatic systems capable of DNA intercalation, aromatic amines and halogenated aromatics are the classic structural alerts, so aromatic ring count and low sp3 fraction matter.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS08	2	TPSA	0.140887	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	topological polar surface area - the size of the polar region	The health-hazard pictogram flags carcinogenicity, mutagenicity, reproductive toxicity, respiratory sensitisation and aspiration hazard. Planar aromatic systems capable of DNA intercalation, aromatic amines and halogenated aromatics are the classic structural alerts, so aromatic ring count and low sp3 fraction matter.	not among the classically expected bulk descriptors; likely acting as a structural marker

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS08	3	NumHDonors	0.121378	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	number of hydrogen-bond donors (OH, NH groups)	The health-hazard pictogram flags carcinogenicity, mutagenicity, reproductive toxicity, respiratory sensitisation and aspiration hazard. Planar aromatic systems capable of DNA intercalation, aromatic amines and halogenated aromatics are the classic structural alerts, so aromatic ring count and low sp3 fraction matter.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS08	4	MACCS_121	0.118889	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	MACCS substructure key 121: matches SMARTS pattern [#7;R]	The health-hazard pictogram flags carcinogenicity, mutagenicity, reproductive toxicity, respiratory sensitisation and aspiration hazard. Planar aromatic systems capable of DNA intercalation, aromatic amines and halogenated aromatics are the classic structural alerts, so aromatic ring count and low sp3 fraction matter.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS08	5	MACCS_119	0.115361	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	MACCS substructure key 119: matches SMARTS pattern [#7]=*	The health-hazard pictogram flags carcinogenicity, mutagenicity, reproductive toxicity, respiratory sensitisation and aspiration hazard. Planar aromatic systems capable of DNA intercalation, aromatic amines and halogenated aromatics are the classic structural alerts, so aromatic ring count and low sp3 fraction matter.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS09	1	MolLogP	0.378431	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	lipophilicity - how strongly the compound prefers oil to water	The environmental pictogram is assigned on aquatic toxicity and persistence. High lipophilicity drives bioaccumulation, and halogenated aromatics resist degradation, so LogP together with halogen-bearing fingerprint bits are the strongest predictors.	yes - this is a descriptor a chemist would expect to matter for this hazard

Pictogram Code	Rank	Feature	Mean Abs SHAP	SHAP Direction	What The Descriptor Measures	Chemical Interpretation	Matches Chemical Expectation
GHS09	2	MACCS_137	0.358769	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	MACCS substructure key 137: matches SMARTS pattern [!C;!c;R]	The environmental pictogram is assigned on aquatic toxicity and persistence. High lipophilicity drives bioaccumulation, and halogenated aromatics resist degradation, so LogP together with halogen-bearing fingerprint bits are the strongest predictors.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS09	3	Chi1	0.30195	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	first-order connectivity index - bond-level branching	The environmental pictogram is assigned on aquatic toxicity and persistence. High lipophilicity drives bioaccumulation, and halogenated aromatics resist degradation, so LogP together with halogen-bearing fingerprint bits are the strongest predictors.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS09	4	LabuteASA	0.25488	positive - a HIGHER value of this descriptor pushes the prediction towards HAZARDOUS	approximate molecular surface area	The environmental pictogram is assigned on aquatic toxicity and persistence. High lipophilicity drives bioaccumulation, and halogenated aromatics resist degradation, so LogP together with halogen-bearing fingerprint bits are the strongest predictors.	not among the classically expected bulk descriptors; likely acting as a structural marker
GHS09	5	MACCS_65	0.129309	negative - a HIGHER value of this descriptor pushes the prediction towards SAFE	MACCS substructure key 65: matches SMARTS pattern c:n	The environmental pictogram is assigned on aquatic toxicity and persistence. High lipophilicity drives bioaccumulation, and halogenated aromatics resist degradation, so LogP together with halogen-bearing fingerprint bits are the strongest predictors.	not among the classically expected bulk descriptors; likely acting as a structural marker

This table has 12 columns in full; 8 are shown here and 4 are omitted so the table stays legible. The complete table, with every column, is sheet 'S4b_SHAP_interpretation' of the accompanying Excel workbook.

45 rows. Source sheet: S4b_SHAP_interpretation.

Table S5

Malaysian industrial validation, per hazard class.

GHS Column	Pictogram Code	Meaning	N True Positive In Set	Accuracy	Precision	Recall	F1	MCC
GHS01_Explosive	GHS01	Explosive (exploding bomb)	0	1.0	0.0	0.0	0.0	
GHS02_Flammable	GHS02	Flammable (flame)	28	0.8837	0.871	0.9643	0.9153	0.7413
GHS03_Oxidising	GHS03	Oxidiser (flame over circle)	2	1.0	1.0	1.0	1.0	1.0
GHS04_CompressedGas	GHS04	Compressed gas (gas cylinder)	10	0.9535	0.8333	1.0	0.9091	0.8848
GHS05_Corrosive	GHS05	Corrosive (corrosion)	14	0.8605	0.8333	0.7143	0.7692	0.6742
GHS06_AcuteToxicity	GHS06	Acute toxicity (skull and crossbones)	16	0.7907	0.6522	0.9375	0.7692	0.6214
GHS07_Irritant	GHS07	Irritant / harmful (exclamation mark)	37	0.8605	0.9429	0.8919	0.9167	0.4974
GHS08_HealthHazard	GHS08	Serious health hazard (health hazard)	36	0.7674	0.8611	0.8611	0.8611	0.1468
GHS09_Environmental	GHS09	Environmental hazard (environment)	25	0.4884	0.8	0.16	0.2667	0.1607

9 rows. Source sheet: S5_malaysia_per_class.

Table S5b

Malaysian industrial validation, per sector.

Sector	N Compounds	Label Accuracy	Hazard Recall	N Actual Hazard Labels	N Hazards Correctly Flagged
Johor 2019 Emergency	12	0.9074	0.8364	55	46
Palm Oil Processing	7	0.9048	0.9	20	18
Petrochemicals	9	0.8765	0.8387	31	26
Rubber Processing	7	0.7619	0.7083	24	17
Semiconductor (Penang)	8	0.7361	0.6579	38	25

5 rows. Source sheet: S5b_malaysia_per_sector.

Table S5c

The chemicals implicated in the 2019 Sungai Kim Kim incident at Pasir Gudang, Johor, with predicted and reference hazard profiles.

Query Name	PubChem Name	CID	MolecularFormula	Sector	Incident Role	Substitution Note
acrylonitrile	Acrylonitrile	7855	C3H3N	Johor 2019 Emergency	principal agent identified by the Ministry of Health	
acrolein	Acrolein	7847	C3H4O	Johor 2019 Emergency	principal agent; a severe respiratory irritant	
benzene	Benzene	241	C6H6	Johor 2019 Emergency	volatile aromatic detected in air and water samples	
toluene	Toluene	1140	C7H8	Johor 2019 Emergency	volatile aromatic detected in air samples	
p-xylene	P-Xylene	7809	C8H10	Johor 2019 Emergency	volatile aromatic detected in air samples	'xylene' is an isomer mixture rather than a single compound; 'p-xylene' used as a representative structure
ethylbenzene	Ethylbenzene	7500	C8H10	Johor 2019 Emergency	volatile aromatic detected in air samples	
hydrogen sulfide	Hydrogen Sulfide	402	H2S	Johor 2019 Emergency	released from the anaerobic river sediment	
methane	Methane	297	CH4	Johor 2019 Emergency	released from the anaerobic river sediment	
methyl mercaptan	Methanethiol	878	CH4S	Johor 2019 Emergency	odorous sulfur compound reported at the site	
d-limonene	(+)-Limonene	440917	C10H16	Johor 2019 Emergency	solvent component reported in the waste mixture	
hydrogen chloride	Hydrochloric Acid	313	ClH	Johor 2019 Emergency	acid gas reported in the waste mixture	
styrene	Styrene	7501	C8H8	Johor 2019 Emergency	monomer reported among the solvent residues	

This table has 38 columns in full; 7 are shown here and 31 are omitted so the table stays legible. The complete table, with every column, is sheet 'S5c_johor_2019' of the accompanying Excel workbook. 12 rows. Source sheet: S5c_johor_2019.

File S1

Complete software environment. Every version is pinned, and a single random seed of 42 is applied throughout, so the analysis reproduces exactly.

```
# =====
# STEP1_environment_requirements.txt
# GHS Chemical Hazard Classification project - Sareer Ahmad
# Generated: 2026-08-06T06:22:08
# Python   : 3.11.15 (D:\GHS_Project\.venv\Scripts\python.exe)
# Platform : Windows-10-10.0.19045-SP0
# Random seed used project-wide: 42
#
# INSTALLATION NOTE (documented deviation, see Rule 2):
#   The official python.org Windows installer (python-3.11.9-amd64.exe)
#   failed twice with WiX bootstrapper exit code 0x3 on this machine
#   (no administrator elevation available for the chained MSI packages).
#   FALLBACK APPLIED: the 'uv' package manager (Astral) was used to
#   install a standalone CPython 3.11.15 build, which requires no
#   Windows installer at all. All packages were then installed with
#   'uv pip install', which is a drop-in replacement for pip.
#   conda was not available on this machine, so the conda fallback
#   named in the proposal could not be used.
# =====

# =====
# PINNED DEPENDENCIES - DO NOT UPGRADE WITHOUT READING THIS
# =====
#
# starlette==0.52.1
#   WHY: Streamlit 1.61.0 declares 'starlette<2,>=0.46.0', but that range is
#   too permissive. Starlette 1.4.0 added a required keyword-only
#   argument 'thread_minimum_size' to GZipResponder.__init__, which
#   Streamlit's own gzip middleware subclasses without passing. With
#   Starlette >= 1.0 installed, every HTTP response from the web
#   application fails with HTTP 500 and the page never loads. Installing
#   with an unconstrained resolver picks the newest Starlette and
#   reproduces the fault, so the pin is required.
#   SYMPTOM IF UNPINNED: TypeError: GZipResponder.__init__() missing 1 required keyword-only argument: 'thread_minimum_size'
#   VERIFIED WORKING: streamlit 1.61.0 + starlette 0.52.1
#
starlette<1.0
# =====

# --- Directly required by the research proposal ---
requests==2.34.2
pandas==3.0.5
numpy==2.4.6
rdkit==2026.03.5
scikit-learn==1.9.0
xgboost==3.2.0
imbalanced-learn==0.14.2
shap==0.51.0
matplotlib==3.11.1
seaborn==0.13.2
```

```
scipy==1.17.1
joblib==1.5.3
streamlit==1.61.0
reportlab==5.0.0
tqdm==4.70.0
pubchempy==1.0.5
lightgbm==4.7.0
iterative-stratification==0.1.9
openpyxl==3.1.5
starlette==0.52.1
```

```
# --- Complete frozen environment (pip freeze) ---
```